Every Constructor an Interaction

Spatial structure made visible to bisimulation in interactive graph-structured lambda theories

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Abstract

Spatial logics separate more than bisimulation does. This is not a defect and it is not news: Caires observed two decades ago that the equivalence induced by a spatial logic for the π-calculus sits strictly between structural congruence and strong bisimilarity, and the intensionality of such logics was settled shortly afterwards by Sangiorgi and by Hirschkoff, Lozes and Sangiorgi. Yet the same observation, restated, is routinely taken to refute any claim that a generated spatial–behavioural logic is *adequate* — that its formulae separate exactly what interaction separates. The two claims look incompatible and are not.

They are not incompatible because bisimilarity is not one relation. It is indexed by the theory that supplies the contexts, and a theory can be enlarged. We give a uniform, purely syntactic enlargement. Given an interactive graph-structured lambda theory $\mathbb{G}$ with interaction constructor K, adjoin for each term constructor C a nullary atom $C_\bullet$, overload K to accept an atom against an argument list, and adjoin the two rewrite rules that build C from its parts and open C into its parts. Nothing else. In the resulting theory $\mathbb{G}^+$ every term former is an interaction, and since bisimulation observes interactions it observes every term former: bisimilarity on $\mathbb{G}^+$ collapses onto the equational theory of $\mathbb{G}$, which is where the spatial connectives already were. The two disputed propositions are then the two endpoints of one monotone family, indexed by how much of the signature one chooses to open.

This is folklore in the spatial logics community. Our contribution is not the phenomenon but its handling: earlier results reach extensionality *semantically*, by exhibiting models — typically distributed calculi with failures — whose observational equivalence happens to be characterised by a spatial logic. We reach it *syntactically*, by a transformation of presentations that is mechanical in the signature and chooses nothing, and therefore *generally*, for any interactive GSLT rather than for a fixed calculus. We record what the construction costs. It is idempotent because the adjoined constructors are nullary; it is calibrated, landing on the equational theory rather than on raw syntax, because the equations are kept as equations; and it is not object-capability safe, since unrestricted opening destroys encapsulation. We describe the shape of the capability discipline that repairs this, observe that its factory is the unique constructor admitting no destructor, and do not develop it here.

Contents

1 Introduction

1.1 The phenomenon

A spatial–behavioural logic for a process calculus carries two layers. The behavioural layer is modal: formulae speak of the steps a term can take. The structural, or spatial, layer speaks of how the term is put together — that it is a parallel composition of two parts with such-and-such properties, that it is a send on a name, that it is empty.

The layers do not have the same separating power, and the direction of the gap surprises people the first time they meet it. Behavioural equivalence is coarse by design: it identifies terms that no experiment can tell apart. Structural predicates are finer, because they can ask questions no experiment can pose. The standard example is as old as CCS. In a calculus with prefix, sum and parallel composition,

$$a.\mathbf{0} \mid b.\mathbf{0} \qquad \text{and} \qquad a.b.\mathbf{0} + b.a.\mathbf{0}$$

are strongly bisimilar — the expansion law is exactly the statement that they are — and they are not structurally congruent. The first satisfies the separating conjunction $\neg\mathbf{0} \mid \neg\mathbf{0}$: it splits into two nonempty parts. The second does not. A logic with a separating conjunction therefore distinguishes two terms that bisimulation identifies.

1.2 The apparent contradiction

Now put that observation next to a second one. When the logic is not designed but *generated* from the presentation of the calculus — one structural connective per term former, one modality per rewrite rule and redex position — one wants an adequacy theorem: logical equivalence coincides with the calculus's observational equivalence. Adequacy is what makes a generated logic worth generating. Too coarse and there are observable distinctions the logic cannot state; too fine and the logic invites its user to chase differences no experiment could settle.

So one wants to assert both of:

$$\text{logical equivalence} \;=\; \text{context-labelled bisimulation}, \tag{A}$$

$$\text{the spatial layer} \;\subsetneq\; \text{context-labelled bisimulation}. \tag{B}$$

Read carelessly these are inconsistent, and a careful reader will say so. We have seen (B) offered as a refutation of (A) more than once, by readers who knew the literature well enough to recognise (B) and not well enough to know what answers it.

1.3 The resolution, in one sentence

Bisimilarity is parameterised by the theory that supplies the contexts. (B) is a statement about bisimilarity in $\mathbb{G}$. (A) is a statement about bisimilarity in a theory $\mathbb{G}^+$ obtained from $\mathbb{G}$ by a construction that makes term formation itself an interaction. Once the parameter is exhibited, (A) and (B) are the two endpoints of a monotone family, and there is nothing to reconcile.

The construction is the subject of §4. Its slogan is:

> *Make every term former an interaction. Bisimulation observes interactions. Therefore bisimulation observes every term former.*

1.4 What is new here, and what is not

The phenomenon is known. Section 2 sets out three lineages of prior work, of which the third — extensionality of spatial observation — already establishes that spatial logics need not be counted intensional. We claim no priority over any of it.

What we add is a change of method, and the change buys generality. The prior extensionality results are *semantic*: they exhibit a calculus, typically a distributed one with local and remote communication and partial failures, and show that its observational equivalence is characterised by a spatial logic. The extra separating power comes from features of that calculus — failures, in particular — which are modelling choices about a particular domain.

Our construction is *syntactic*. It is a transformation of presentations. It reads the signature, adjoins one nullary atom per constructor and two rules per constructor, and stops. It chooses nothing, it is defined for any interactive GSLT rather than for one calculus, and because it is mechanical in the presentation it composes with the machinery that generates the logic from the same presentation. That last point is the one that matters for our purposes: if both the logic and the enlargement are functions of the signature, then adequacy is the statement that two functions of the same argument agree, rather than a coincidence to be checked calculus by calculus.

We also record three things about the construction that we have not seen written down: that it is idempotent, and why (§5.4); that its calibration — landing on the equational theory rather than below it — depends on a presentational choice that is easy to get wrong (§5.1); and that it is not object-capability safe (§7).

1.5 What this note does not do

It does not develop the capability discipline of §7. It does not treat weak equivalences, quantitative refinements, or the interaction with cost accounting beyond the sketch of §6.3. And it makes no claim that opening a term is a sensible thing to permit in a deployed system. The enlarged theory is a measuring instrument. It exists to say what bisimulation *can in principle* see; it is not a habitat.

2 Three lineages

The literature that bears on this contains three distinguishable strands, and much of the confusion we are addressing comes from reading them as one. We set them out in the order they arose.

The observation. Caires, studying an expressive spatial logic for the synchronous π-calculus with recursion, gives coinductive and equational characterisations of the equivalence the logic induces and locates it: strictly between structural congruence and strong bisimilarity [1]. The companion development with Cardelli [2, 3] is explicit that formulae are taken to denote processes up to structural equivalence rather than up to bisimulation, precisely because a tensor operator that distinguishes bisimilar processes is difficult to reconcile with the latter. This is (B), stated by the people who first had to confront it.

The intensionality results. Sangiorgi [4] and then Hirschkoff, Lozes and Sangiorgi [5, 6] settle where the induced equivalence sits: for logics of this kind it is essentially structural congruence. Lozes adds the elimination results — that in the static ambient logic the adjuncts contribute no expressive power [7], later strengthened by Dawar, Gardner and Ghelli through model-comparison games [10] — and the picture is completed by Caires and Lozes [9], who show that a minimal dynamic spatial logic already encodes action quantification and action modalities, so that the spatial and modal fragments have the same separating power, coinciding with structural congruence modulo permutation of actions. Lozes' thesis [8] is the systematic account.

This strand says the logic is *intensional*: it sees syntax. It is often read, wrongly, as saying that nothing can be done about this.

The extensionality results. Something can be done about it. Caires and Vieira [11] address the tension directly, showing that there are natural models in which extensional observational equivalences *are* characterised by spatial logics, composition and void included — and concluding that spatial observations need not always be counted intensional, even when expressive enough to describe structure. Their development is carried out in a minimalist distributed calculus with local synchronous communication, local computation, asynchronous remote communication, and partial failures. The additional separating power available to an observer in that setting is what closes the gap. A related model-theoretic treatment is given by Tuosto and Vieira [12]; Caires' tutorial survey [13] is the best single entry point to the area.

strand	question settled	what it does not settle
the observation	where the induced equivalence sits relative to $\sim$ and $\equiv$	whether the gap is intrinsic
intensionality	that the induced equivalence is essentially $\equiv$; which connectives are eliminable	whether an observer could be given the missing power
extensionality	that in suitable models observational equivalence is characterised spatially	whether this can be obtained uniformly from a presentation

Table 1: The three strands, and the question each leaves open. The present note addresses the right-hand cell of the third row.

Table 1 is the summary we wish had been available to us. A reader who knows only the second strand will conclude that spatial logics are irreducibly intensional and that any adequacy claim about a generated logic is confused. That conclusion does not follow, and the third strand is where the answer lives.

3 Preliminaries

We recall only what is used. For graph-structured lambda theories in general we refer to [14].

Definition 1 (GSLT)**.** A *graph-structured lambda theory* is a triple $\mathbb{G} = (\Sigma, E, R)$ where Σ is the signature of a lambda theory — so constructors may bind, and terms may contain variables — E is a set of equations over Σ-terms, and R is a set of rewrite rules, the graph of rewrites being adjoined to the structure rather than the structure being enriched over graphs. We write $\mathcal{T}(\Sigma)$ for the terms and $=_E$ for the least congruence containing E.

Definition 2 (Interactive)**.** $\mathbb{G}$ is *interactive* when Σ carries a distinguished binary constructor K, the *interaction constructor* or *cut*, such that the left-hand side of every rule in R is of the form $\mathsf{K}(l_1, l_2)$ up to $=_E$.

Parallel composition plays this role for CCS, the π-calculus, the ambient calculus and the rho calculus; application plays it for the lambda calculus.

Definition 3 (Context-labelled transitions)**.** For $P, P' \in \mathcal{T}(\Sigma)$ write $P \xrightarrow{(\rho,D)} P'$ when there are a rule $\rho = (l \to r) \in R$, a one-hole context $D[-]$ and a substitution σ with $P =_E D[l\sigma]$ and $P' =_E D[r\sigma]$. The label records both the rule and the position. *Context-labelled bisimulation* $\sim_{\mathbb{G}}$ is the largest symmetric relation closed under matching of these labelled steps.

Remark 4 (Notation)**.** The scientist note [15] writes K_j for what we here call D, and K for the cut. The coincidence of letters there is not accidental — for the rho calculus the cut and the redex context are built from the same constructor — but it is confusing in a general setting, so we separate them: K is a constructor, D is a context.

Definition 5 (The generated logic)**.** $\mathcal{L}(\mathbb{G})$ has formulae

$$\varphi \ ::= \ \top \ \mid \ \neg\varphi \ \mid \ \varphi \wedge \psi \ \mid \ C(\varphi_1, \ldots, \varphi_n) \ \mid \ \langle \rho, D \rangle \varphi$$

with one *structural* connective $C(-,\dots,-)$ per n-ary constructor $C \in \Sigma$ and one *behavioural* modality per rule and redex position. Satisfaction for the structural connectives is

$$P \models C(\varphi_1,\dots,\varphi_n) \iff P =_E C(t_1,\dots,t_n) \text{ with } t_i \models \varphi_i \text{ for each } i,$$

and for the modalities $P \models \langle \rho, D\rangle\varphi$ iff $P \xrightarrow{(\rho,D)} P'$ with $P' \models \varphi$. We write $=_{\mathcal{L}(\mathbb{G})}$ for logical equivalence.

Because K is typically subject to associativity, commutativity and a unit in E, the connective $\mathsf{K}(\varphi,\psi)$ is a separating conjunction: it holds of P exactly when P admits *some* decomposition into two parts satisfying φ and ψ respectively. The existential over decompositions is the whole difficulty. It is not a step, so no modality expresses it, and that is where (B) comes from.

Remark 6 (Which connectives are admitted)**.** Not every generated logic is adequate for anything, and users of a generator have good reason to restrict which connectives are emitted — typically to obtain a fragment whose model checking terminates [14]. We take the full structural layer as the default and return to the restricted case in §6.2, where it becomes a parameter rather than a caveat.

4 The construction

Fix an interactive GSLT $\mathbb{G} = (\Sigma, E, R)$ with cut K, and assume $\mathbb{G}$ is Turing complete.

4.1 Lists, and why Turing completeness is enough for them

K is binary; constructors have arbitrary arity. To present the arguments of an n-ary constructor to K we need to tuple them, so we assume list formers nil and cons with the evident equations, writing $[t_1,\dots,t_n]$ for the obvious term.

Remark 7 (The lists are not an import)**.** Turing completeness supplies them: any of the standard encodings will do, and which one is chosen does not affect anything below, since all that is used is that $[\,\cdot\,]$ is injective up to $=_E$ and that its components are recoverable by the same construction we are about to define. This is worth saying because it is the *only* thing Turing completeness is used for in this note, and it is worth remembering because in §7 we will need something that Turing completeness does *not* supply.

4.2 Atoms, overloading, and two rules per constructor

Definition 8 (The canonical extension)**.** The *canonical extension* $\mathbb{G}^+ = (\Sigma^+, E, R^+)$ of $\mathbb{G}$ is given by:

- $\Sigma^+ = \Sigma \uplus \{\, C_\bullet : C \in \Sigma \,\}$, where each $C_\bullet$ is *nullary*;
- K is overloaded so that $\mathsf{K}(C_\bullet, [t_1,\dots,t_n])$ is well formed whenever C is n-ary;
- E is unchanged;
- $R^+ = R \cup \{\, \mathrm{build}_C, \mathrm{open}_C : C \in \Sigma \,\}$ where, for C of arity n,

$$\begin{aligned} \mathrm{build}_C :&\quad \mathsf{K}(C_\bullet, [x_1,\dots,x_n]) \longrightarrow C(x_1,\dots,x_n), \\ \mathrm{open}_C :&\quad C(x_1,\dots,x_n) \longrightarrow \mathsf{K}(C_\bullet, [x_1,\dots,x_n]). \end{aligned}$$

We call the rules of R *proper* and those of $R^+ \setminus R$ *administrative.*

Three comments on the shape of this definition.

First, the overloading of K rather than the introduction of a fresh application former is the point of the whole construction. K is not an arbitrary constructor; it is the one the transition relation is indexed on, the one that defines what it means for two things to meet. Putting $C_\bullet$ into interaction position re-expresses construction as communication, and communication is precisely what a context-labelled transition system is built to observe. Nothing is being added to the observational apparatus. Syntax is being moved into the part of the theory the apparatus already covers.

Second, build_C alone would be a term *builder*, and a builder reads nothing it did not build: $C_\bullet$ placed beside an opaque P fires only if P is already a list of the right length. It is open_C that makes structure legible, since it applies at any redex position and therefore to terms nobody holds.

Third, open_C and build_C are converse. Neither the theory nor any term loses anything by their presence; what is gained is that the syntax tree of a term becomes a reachable part of its transition behaviour.

Remark 9 (The equation-free presentation)**.** One may prefer a presentation with no equations at all — for instance to hand the theory to a tool that does only rewriting. Then orient each equation of E in both directions, adding both to the administrative rules. The two presentations agree up to administrative steps. We keep E as equations because doing so makes the calibration of §5.1 immediate rather than a proof obligation, and because it localises the source of nondeterminism: with E retained, the branching of open_C comes from E-matching, which is exactly the existential over decompositions that the separating conjunction quantifies. It is worth stressing that these must be *pairs.* Orienting the equations in one direction only would let bisimilarity fall below $=_E$ onto raw syntax, which overshoots: the structural connectives respect $=_E$, so a relation finer than $=_E$ is finer than the logic and adequacy fails from the other side.

Proposition 10 ($\mathbb{G}^+$ is an interactive GSLT)**.** *$\mathbb{G}^+$ is a GSLT, is interactive with the same cut K, and the inclusion $\mathcal{T}(\Sigma) \hookrightarrow \mathcal{T}(\Sigma^+)$ reflects proper transitions: for $P, P' \in \mathcal{T}(\Sigma)$ and $\rho \in R$, $P \xrightarrow{(\rho,D)} P'$ in $\mathbb{G}$ iff the same holds in $\mathbb{G}^+$.*

Proof. Every rule of R^+ has a left-hand side of the form $\mathsf{K}(-,-)$ up to $=_E$: those of R by hypothesis, build_C by construction, and open_C because $C(x_1,\ldots,x_n) =_E \mathsf{K}(C(x_1,\ldots,x_n),u)$ for the unit u of K when K has one. Where K has no unit, replace open_C by $\mathsf{K}(\mathrm{Op}_\bullet,[C(x_1,\ldots,x_n)]) \to \mathsf{K}(C_\bullet,[x_1,\ldots,x_n])$ with $\mathrm{Op}_\bullet$ a further nullary atom; nothing below changes. Reflection of proper transitions holds because $R^+ \supseteq R$ and the added constructors are not mentioned in any rule of R, so no R-redex is created or destroyed by their presence. □

5 Basic properties

5.1 Calibration

Lemma 11 (Calibration)**.** *If $P =_E Q$ then $P \sim_{\mathbb{G}^+} Q$.*

Proof. Transitions are defined up to $=_E$, so $=_E$-equal terms have identical labelled transition sets. □

Trivial as stated, and stated anyway, because it is the property that the alternative presentation of Remark 9 has to work for, and because it is the half of the calibration that fixes the *lower* bound: the enlargement must not push bisimilarity below the equational theory, or it overshoots the logic.

5.2 Exposure

Lemma 12 (Exposure)**.** *Let $P \in \mathcal{T}(\Sigma^+)$ and let C be n-ary. Then*

$$P \xrightarrow{(\mathrm{open}_C, [-])} \mathsf{K}(C_\bullet, [t_1, \ldots, t_n]) \quad \textit{iff} \quad P =_E C(t_1, \ldots, t_n).$$

Consequently the set of open*-transitions available to P at the empty context enumerates exactly the top-level E-decompositions of P, one transition per decomposition, with the atom in the label recording which constructor.*

Proof. Immediate from Definition 8 and the definition of context-labelled transitions, which matches redexes up to $=_E$. □

This is the lemma that does the work, and it is worth pausing on what it says in the case that motivated the whole difficulty. Take $C = \mathsf{K}$ with E containing associativity, commutativity and the unit. Then P's open_K-transitions at the empty context are in bijection with the ways of writing P as $Q \mid R$ — including $P \mid \mathbf{0}$ and $\mathbf{0} \mid P$, and every rebracketing and permutation of a parallel product. The existential over splits in $P \models \varphi \mid \psi$ has become an existential over transitions. That is the entire content of the construction, and everything else is bookkeeping.

5.3 The reconstruction theorem

Theorem 13 (Reconstruction)**.** *For closed $P, Q \in \mathcal{T}(\Sigma)$,*

$$P \sim_{\mathbb{G}^+} Q \iff P =_E Q.$$

Proof. ($\Leftarrow$) is Lemma 11.

($\Rightarrow$) By induction on the size of P, where size is the number of constructor occurrences in some (hence any, since E is size-respecting on the presentations we consider) $=_E$-representative of least size.

Suppose $P \sim_{\mathbb{G}^+} Q$ and $P =_E C(t_1, \ldots, t_n)$ for some $C \in \Sigma$. By Lemma 12, $P \xrightarrow{(\mathrm{open}_C, [-])} \mathsf{K}(C_\bullet, [t_1, \ldots, t_n])$. Since $P \sim_{\mathbb{G}^+} Q$, the term Q must match that step with the *same* label, so open_C is the rule and the context is empty; by Lemma 12 again, $Q =_E C(s_1, \ldots, s_n)$ for some s_i, and

$$\mathsf{K}(C_\bullet, [t_1, \ldots, t_n]) \ \sim_{\mathbb{G}^+} \ \mathsf{K}(C_\bullet, [s_1, \ldots, s_n]).$$

The atom $C_\bullet$ appearing in the label is what makes the constructors agree: distinct constructors give distinct atoms and hence distinct labels, so no rule other than open_C could have matched.

It remains to descend into the arguments. The list $[t_1, \ldots, t_n]$ is itself built from cons and nil, so Lemma 12 applies again at the contexts $\mathsf{K}(C_\bullet, -)$ and its iterates. Matching those steps pairwise yields $t_i \sim_{\mathbb{G}^+} s_i$ for each i, because the labels record position and therefore align the arguments in order. Each t_i is smaller than P, so the induction hypothesis gives $t_i =_E s_i$, whence $P =_E C(t_1, \ldots, t_n) =_E C(s_1, \ldots, s_n) =_E Q$. □

Remark 14 (Where the lambda theory is used)**.** If C binds — an input prefix, an abstraction — then opening it exposes an argument that is not closed. This is the one place where it matters that a GSLT is built on a lambda *theory* rather than on a first-order signature: the argument of a binding constructor is an abstraction, which is a term of the theory, and the induction goes through on abstractions as it does on closed terms. For a first-order signature the construction is more elementary and the restriction to closed terms in Theorem 13 may be dropped.

Remark 15 (Infinite branching)**.** A term with replication has infinitely many E-decompositions and therefore infinitely many $\mathrm{open}_{\mathsf{K}}$-transitions. Nothing in the proof requires image-finiteness; bisimulation is defined for arbitrary branching and the induction is on term size, not on transition depth.

5.4 Idempotence

A referee's next move, on seeing Theorem 13, is to run the construction again. $\mathbb{G}^+$ has term formers that $\mathbb{G}$ did not; generate the logic over $\mathbb{G}^+$ and there will be structural connectives for them; are those not again finer than $\sim_{\mathbb{G}^+}$, and is one not on a ladder with no top?

Proposition 16 (Idempotence)**.** *$\sim_{\mathbb{G}^{++}} = \sim_{\mathbb{G}^+}$, and $=_{\mathcal{L}(\mathbb{G}^+)}$ adds no separating power over $=_{\mathcal{L}(\mathbb{G})}$ on $\mathcal{T}(\Sigma)$.*

Proof. The constructors adjoined by Definition 8 are nullary. A nullary constructor has no arguments to conceal: its structural connective is the atomic proposition "is $C_\bullet$", and by Lemma 12 that proposition is already determined by the build_C-transitions available at the term, hence already behavioural in $\mathbb{G}^+$. Adjoining atoms and rules for the atoms therefore yields nothing not already present, and by Theorem 13 both sides equal $=_E$. □

This is not a side remark. It is the reason $(-)^+$ is a construction rather than the first stage of a colimit, and it is a consequence of a design decision that could easily have gone the other way: had we introduced a family of n-ary destructors, one per constructor, rather than nullary atoms plus an overloaded cut, the ladder would be real.

6 Consequences

6.1 Adequacy

We now import the second strand of §2.

Proposition 17 (Intensionality; known)**.** *For the spatial logics considered in [4, 5, 6, 9], logical equivalence coincides with structural congruence, possibly modulo a permutation of actions. In the present notation: $=_{\mathcal{L}(\mathbb{G})} = =_E$, whenever the target logic specification carries enough boolean structure to form characteristic formulae.*

We state this as an import rather than prove it. The hypothesis is not decorative: a generated logic whose specification omits negation, or omits conjunction, will not separate to $=_E$, and adequacy will fail for reasons that have nothing to do with the present construction [14].

Corollary 18 (Adequacy in the extension)**.** *Under the hypothesis of Proposition 17, for closed Σ-terms*

$$=_{\mathcal{L}(\mathbb{G})} = =_E = \sim_{\mathbb{G}^+} .$$

That is: the logic generated from $\mathbb{G}$ is adequate for context-labelled bisimulation in $\mathbb{G}^+$.

Corollary 19 (Strictness, preserved)**.** $\sim_{\mathbb{G}^+} \subsetneq \sim_{\mathbb{G}}$ *in general, so the spatial layer does strictly refine bisimulation in* $\mathbb{G}$*.*

Example 20 (The witness)**.** In CCS, $P = a.\mathbf{0} \mid b.\mathbf{0}$ and $Q = a.b.\mathbf{0} + b.a.\mathbf{0}$ satisfy $P \sim_{\mathbb{G}} Q$ by the expansion law. In $\mathbb{G}^+$ they are separated at the first step: P has an $\mathrm{open}_{\mathsf{K}}$-transition at the empty context to $\mathsf{K}(\mathsf{K}_\bullet, [a.\mathbf{0}, b.\mathbf{0}])$, and the only open-transition Q has at the empty context is open_+, carrying a different atom in its label. Correspondingly $P \models \neg\mathbf{0} \mid \neg\mathbf{0}$ and $Q \not\models \neg\mathbf{0} \mid \neg\mathbf{0}$.

Corollaries 18 and 19 are (A) and (B). They are both true, of different relations, and the difference is a parameter that was suppressed rather than a disagreement.

6.2 The dial

Nothing forces the construction to be run on the whole signature.

Definition 21 (Partial extension)**.** For $\mathcal{C} \subseteq \Sigma$ let $\mathbb{G}^{+\mathcal{C}}$ be Definition 8 with atoms and rules adjoined only for $C \in \mathcal{C}$.

Proposition 22 (Monotonicity)**.** *If* $\mathcal{C} \subseteq \mathcal{C}'$ *then* $\sim_{\mathbb{G}^{+\mathcal{C}'}} \subseteq \sim_{\mathbb{G}^{+\mathcal{C}}}$*, with* $\sim_{\mathbb{G}^{+\varnothing}} = \sim_{\mathbb{G}}$ *and* $\sim_{\mathbb{G}^{+\Sigma}} = =_E$*.*

Proof. Adding rules only adds transitions that must be matched, so the largest bisimulation can only shrink. The endpoints are Proposition 10 and Theorem 13. □

Corollary 23 (Relativised adequacy)**.** *Let* $\mathcal{L}_{\mathcal{C}}(\mathbb{G})$ *be the fragment of the generated logic whose structural connectives are those of* $\mathcal{C}$*. Then* $=_{\mathcal{L}_{\mathcal{C}}(\mathbb{G})}$ *is adequate for* $\sim_{\mathbb{G}^{+\mathcal{C}}}$ *under the hypothesis of Proposition 17 relativised to* $\mathcal{C}$*.*

This is the form in which the result is most useful, and it puts the earlier caveat about restricted connective sets in its proper place. A user who dials the generator down to a finitely checkable fragment is not thereby abandoning adequacy; they are moving along the family of Proposition 22, and adequacy holds at every point of it, against the bisimulation of the correspondingly partial extension. The logic and the enlargement are indexed by the same parameter, and adequacy is the statement that they agree.

A second dial gives the same family without touching the signature: fix $\mathbb{G}^{+\Sigma}$ and vary instead which administrative labels an observer is permitted to see. Observing none recovers $\sim_{\mathbb{G}}$; observing all recovers $=_E$. Whether the two dials give the same family in general we have not checked.

6.3 Cost

We record a direction rather than a result. In a cost-accounted GSLT [17] every rewrite carries a price, and the administrative rules are rewrites like any other. Two consequences suggest themselves.

Evaluating a structural formula of depth d against a term corresponds to a path of d nested open-steps. So the price of a structural observation, which in [15] is stipulated as a monotone schedule in formula depth, is on this reading a *step count* — monotone in depth because depth is what it counts. And the separating conjunction is expensive for a reason that is now visible rather than asserted: confirming $\varphi \mid \psi$ requires searching the E-decompositions, which is a walk over administrative transitions rather than a single step.

Whether the induced schedule agrees with the stipulated one, and what the exchange rate between proper and administrative steps ought to be, we leave open.

7 The construction is not capability-safe

We state this plainly rather than hedging it, because a reader who knows object-capability discipline will see it in the first line of Definition 8 and will want to know whether we noticed.

Unrestricted open destroys encapsulation. The rule applies at every redex position, so any subterm of any term, held by anybody or by nobody, may be opened; and opening yields the arguments as data. In a calculus where an opaque reference is the unit of authority, this is fatal: open the reference and harvest whatever authority its contents carry. Confinement fails, and with it every argument that depends on it.

The repair is known in outline. Mike Stay's suggestion is that constructors and destructors should each take a capability token, so that the right to build and the right to open are held rather than ambient. There are two readings of such a token — a *permission*, licensing the rule, or a *brand*, so that build_C stamps the term and open_C checks the stamp — and the second is the more useful, since it makes legibility a property of the term rather than a global property of the world, which is what any application to distributed systems will want. Note that the token must gate the *pair*: the calibration of Lemma 11, in the equation-free presentation of Remark 9, requires that an observer able to take an administrative step is able to take its converse, and an asymmetric gate breaks that.

Tokens require a factory: a constructor that manufactures them. Two observations, and then we stop.

Remark 24 (The factory is an import)**.** Remark 7 used Turing completeness to obtain lists. It cannot be used again here. Unforgeability is not a claim about what can be computed but about what a context can *write down*, and in a signature of free constructors every term is writable by anybody. So a genuine source of privacy must be adjoined — restriction, in the general case, giving a factory of the shape $\mathsf{let}\ c = \mathsf{freshCap}()\ \mathsf{in}\ P$, which is essentially $(\nu c)P$. Reflective calculi, in which a name is a quoted process, may be able to discharge the import internally; whether quotation *originates* privacy or merely propagates it from a seed is a question we do not settle here.

Remark 25 (The factory is the unique constructor with no destructor)**.** Whatever the factory is, it is the one constructor to which the construction of Definition 8 must *not* be applied. Give it an open rule and a token can be opened; if a token can be opened it can be rebuilt by build; if it can be rebuilt it can be forged; and if it can be forged nothing is a capability and the gate collapses. So the capability-safe version of the construction is not "the canonical extension, with a caveat" but "the canonical extension, with exactly one exception, and the exception is what makes the rest of it safe". We think this is the right way to state unforgeability in this setting: a capability is the output of the unique destructor-free constructor.

Beyond this the design has real content — whether the token is conserved or purchased, whether it is consumed by use or held, how it interacts with a cost-accounting resource signature — and we do not pursue it. It is worth being clear about the scope of the omission. Nothing in §§4–6 depends on it, because $\mathbb{G}^+$ is used there as an instrument for locating an equivalence, not as a calculus in which anything runs. The moment one wants to *execute* in $\mathbb{G}^+$, the capability discipline stops being optional.

8 What this settles, and for whom

We collect the consequences for the framework this note was written to serve.

Adequacy claims survive contact with intensionality results. A generated spatial–behavioural logic may be asserted adequate, provided the assertion names the relation: bisimulation in the extension, not in the base. Assertions that omit the index are not wrong so much as ambiguous, and the ambiguity is what invites the objection.

Structural observation is not a second channel. An architecture that lists structural inspection alongside interaction as a separate mode of access is, on this reading, double-counting. Structural inspection *is* interaction — with administrative redexes — and what distinguishes the modes is not which channel they use but what they cost and what they destroy. That is a better distinction, since it is the one that does work downstream.

Spatial and behavioural addresses coincide. In $\mathbb{G}^+$ a decomposition site is a redex position. Constructions that need to identify a structural locus with a modal label — and there are several in [15] — are therefore well typed rather than analogical. This does not by itself supply an enabling theorem: a syntactic locus that is a redex position in $\mathbb{G}^+$ need not correspond to a *proper* step that is reachable in $\mathbb{G}$, and claims about reachability still require their own argument.

Morphisms. The category **GSLT** has bisimulation-preserving maps as morphisms [14], and an assignment that depends on spatial structure is not functorial over them; the usual repair is to narrow the morphism class by hand to maps preserving the generated logic. Under Theorem 13 the narrowing is not a narrowing: logic-preserving maps of $\mathbb{G}$ are exactly bisimulation-preserving maps of $\mathbb{G}^+$. The repair becomes functoriality on the image of $(-)^+$.

An in-principle ceiling. Adequacy in the extension is a claim about what an ideal observer could separate, not about what any particular agent can afford. Arguments that move from adequacy to a conclusion about an actual learner need the affordability premise stated separately; the fragment an agent occupies is fixed by its budget, and §6.2 is where the two meet.

9 Open problems

1. **Canonicity as an adjunction.** We have called $(-)^+$ canonical on the grounds that it is mechanical in the presentation and chooses nothing. Is it a left adjoint — free on $\mathbb{G}$ among extensions making $\mathcal{L}(\mathbb{G})$ behavioural, hence initial among them? An adjunction would upgrade "evident construction" to "forced construction", which matters wherever the absence of a design choice is doing argumentative work.

2. **Do the two dials agree?** §6.2 gives a family indexed by a subset of the signature and a second indexed by a set of observable administrative labels. Are they the same family?

3. **The capability-indexed family.** With Stay's tokens the relation $\sim_{\mathbb{G}^+}$ becomes a family indexed by the tokens an observer holds, interpolating between $\sim_{\mathbb{G}}$ and $=_E$ with the index an object of the calculus rather than a parameter of the metatheory. This is the sharpest form of the resolution and we have not developed it.

4. **No capability amplification.** The security claim wants a theorem: one cannot open one's way to a token one was not given. The statement has a circular smell — unforgeability rests on the gate, the gate rests on tokens — and needs a well-foundedness argument.

5. **Derived pricing.** Does the step count of §6.3 agree with the stipulated schedules used in cost-accounted models, and under what exchange rate?

6. **Weak equivalences.** Everything here is strong. Administrative steps are natural candidates for being made silent, and the resulting weak theory is presumably where the connection back to $\sim_{\mathbb{G}}$ is cleanest.

7. **Relation to the semantic route.** [11] obtains extensionality from partial failures. Is there a translation carrying that model into a partial extension of the present kind, so that failures and administrative openings turn out to be two presentations of one mechanism?

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